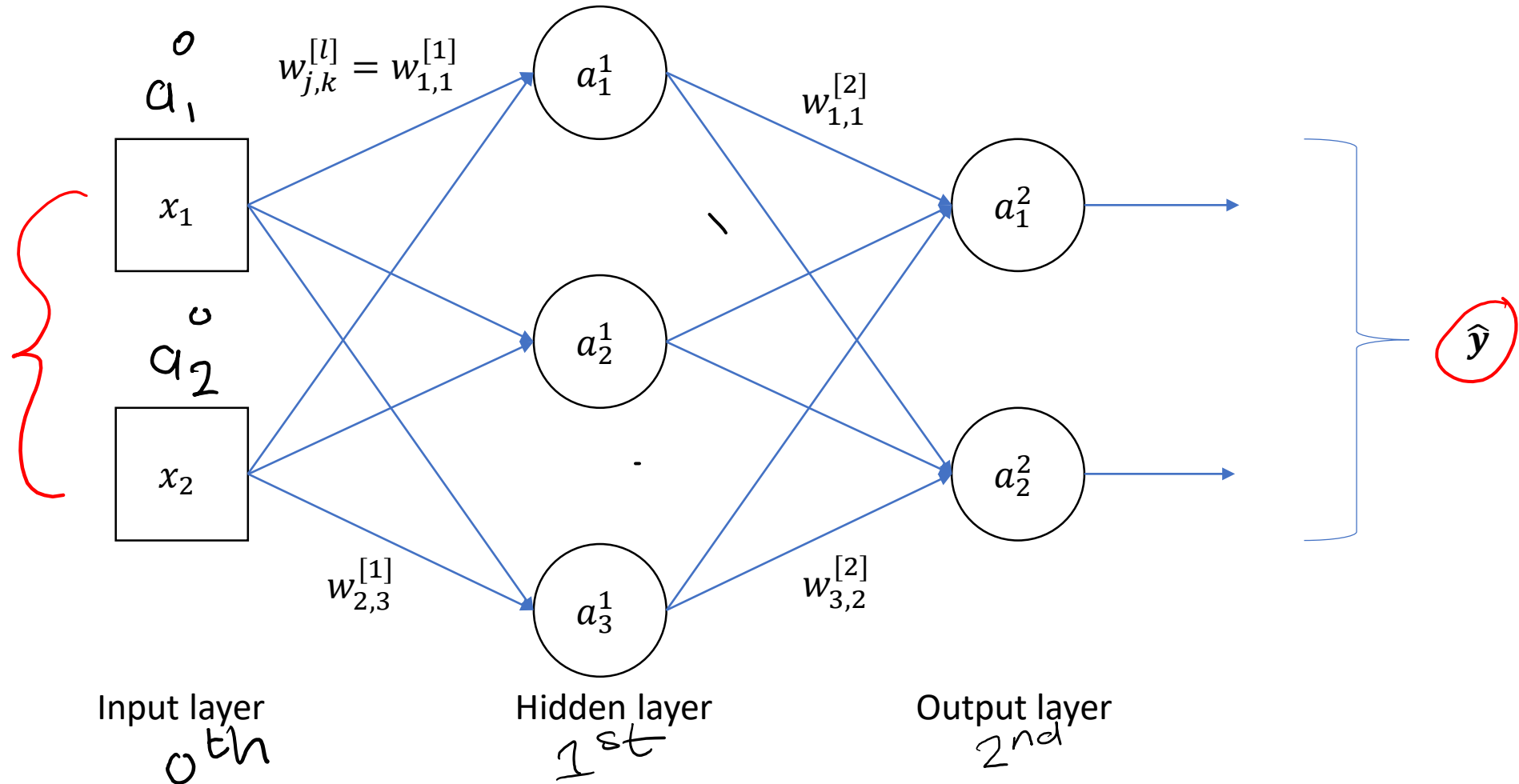


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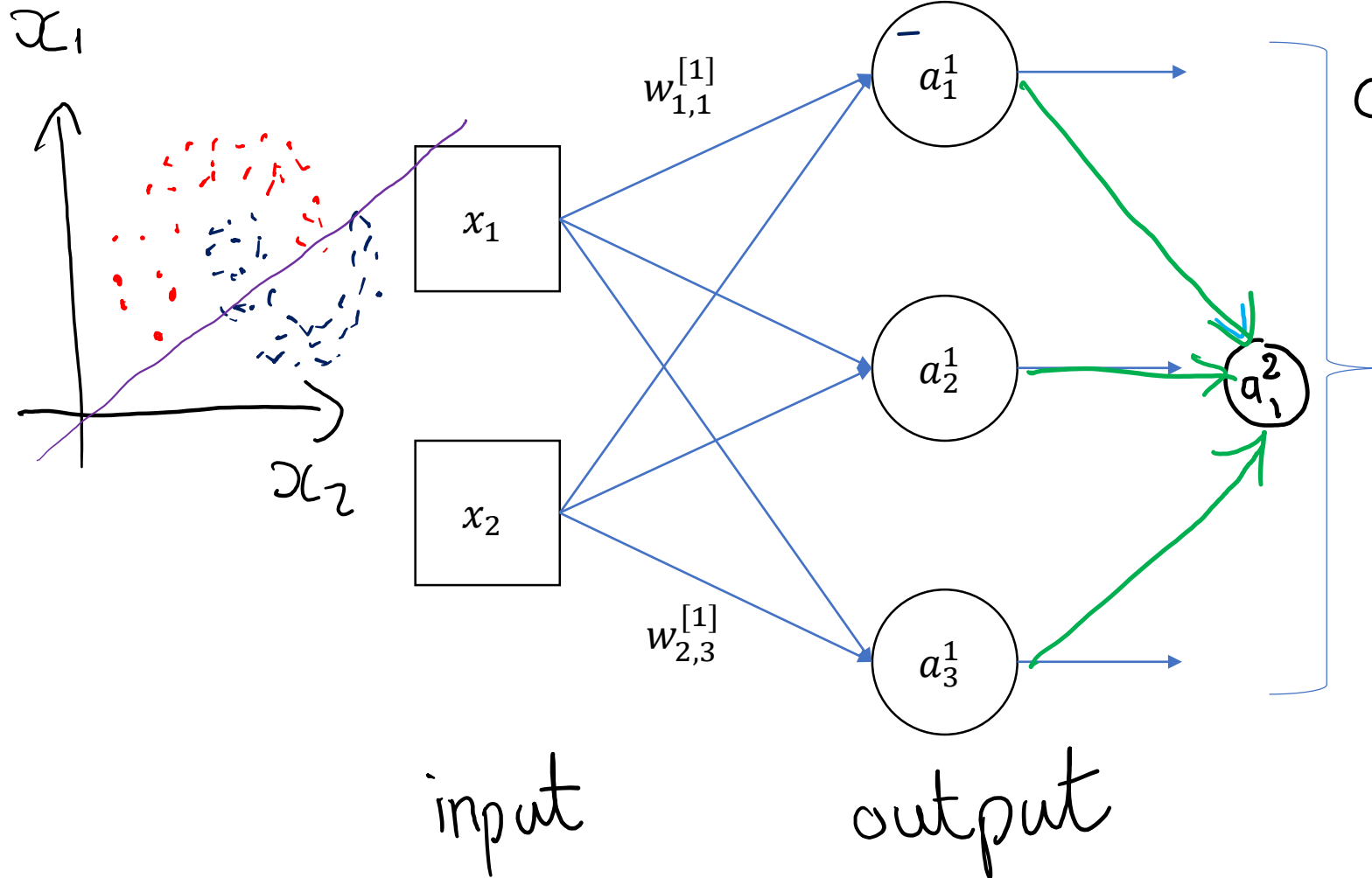
Fully Connected, Multi-layer, feed-forward neural networks

Multi-layer Neural Networks



Fully connected/Dense network: Each neuron in a given layer is connected to every neuron in the next layer

Why do we need depth?



$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

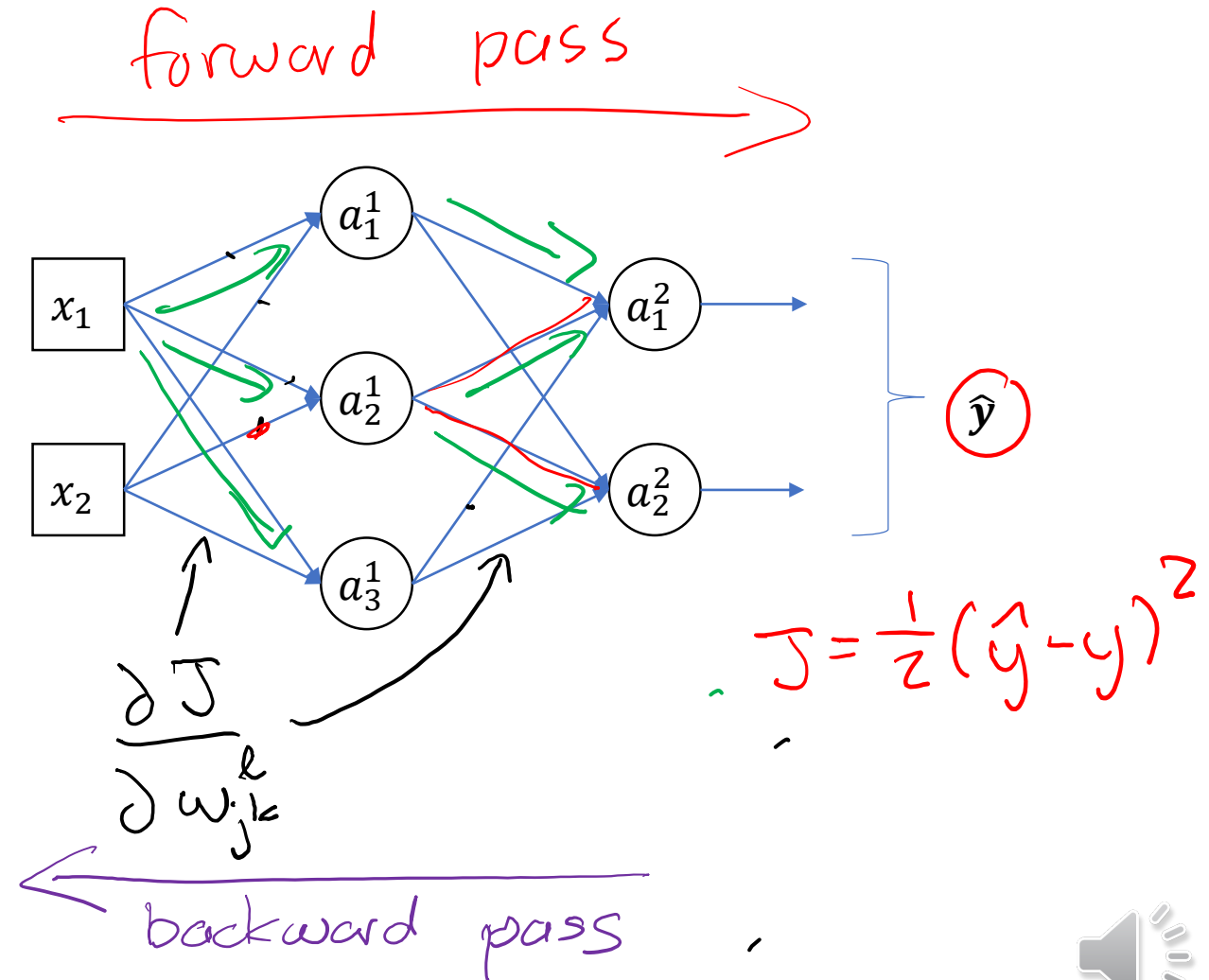
$$a_1 = \frac{1}{1 + e^{-(\underbrace{w_{11}x_1 + w_{21}x_2}_{\text{linear combination of } x})}}$$

$$\hat{y} = a_1^2 = \frac{1}{1 + e^{-(w_{11}^2 \underbrace{\sigma(w_{11}x_1 + w_{12}x_2) + \dots}_{a_1})}}$$



Training operation

1. Provide data to the input
 - Propagate activations forward to generate output
2. Determine error using objective function
3. Optimize weights w.r.t objective
 - Gradient descent methods
 - Use backpropagation to determine gradients



Next up:

The forward pass

